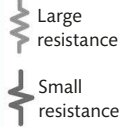




Forming memory circuits

Modeling digital electronic circuits on the brain means storing and recalling information. In the brain, remembering involves the repeated use of particular pathways between neurons that strengthen their junctions (synapses) to form a “memory circuit.” In electronics, a device in development known as the memory resistor or memristor offers a similar function.

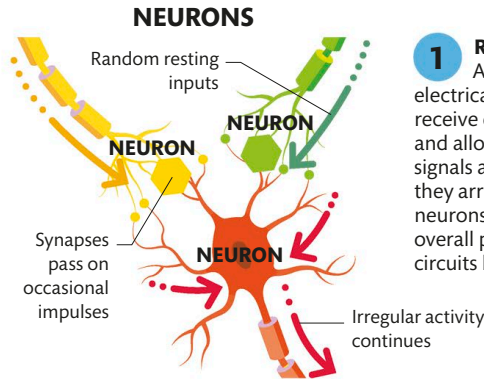
KEY



IN 2019, AN
AI PROGRAM
CALLED **PLURIBUS**
BEAT 5 ELITE HUMAN
POKER PLAYERS

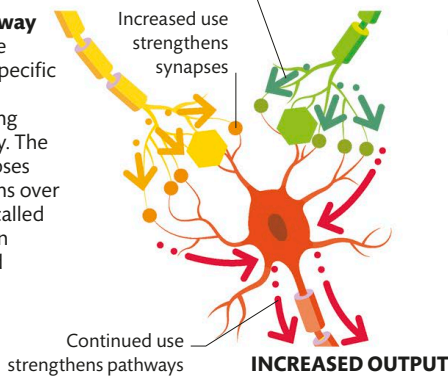


1 Resting state
Nerve impulses pass randomly between a group of neurons—only three are shown here, but there could be thousands. Some synaptic junctions send them on easily, others less so. There is no overall pattern and no definite outcome.

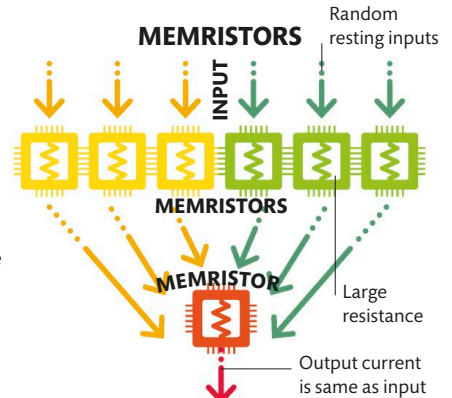


Organized, more frequent inputs

2 Memory pathway
Recurring, more frequent impulses in specific patterns represent a movement or fact being committed to memory. The repeatedly used synapses boost their connections over time, a characteristic called long-term potentiation (LTP; see pp.26–27 and pp.136–137).

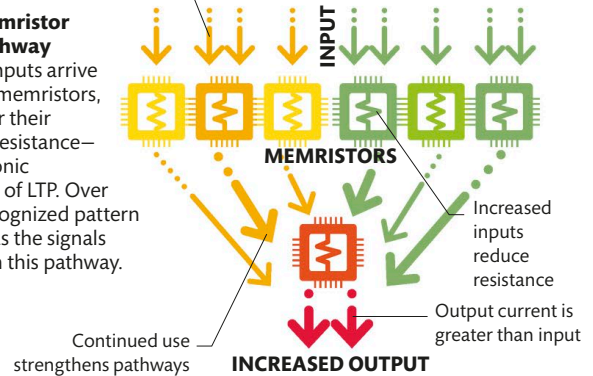


1 Resting state
A set of electrical memristors receive equal inputs and allow through signals as and when they arrive. Like the neurons, there is no overall pattern and the circuits hardly change.



Organized inputs

2 Memristor pathway
Stronger inputs arrive at certain memristors, which alter their electrical resistance—the electronic equivalent of LTP. Over time, a recognized pattern develops as the signals strengthen this pathway.



ELECTRONIC TELEPATHY

Telepathy is the hypothetical direct communication between brains, bypassing senses such as sight. In an experiment using a block-based computer game, instructions to rotate blocks were collected from two players' brains in the form of EEG readings and then communicated, via a transcranial magnetic stimulation (TMS) cap, to a third player to make the moves.

